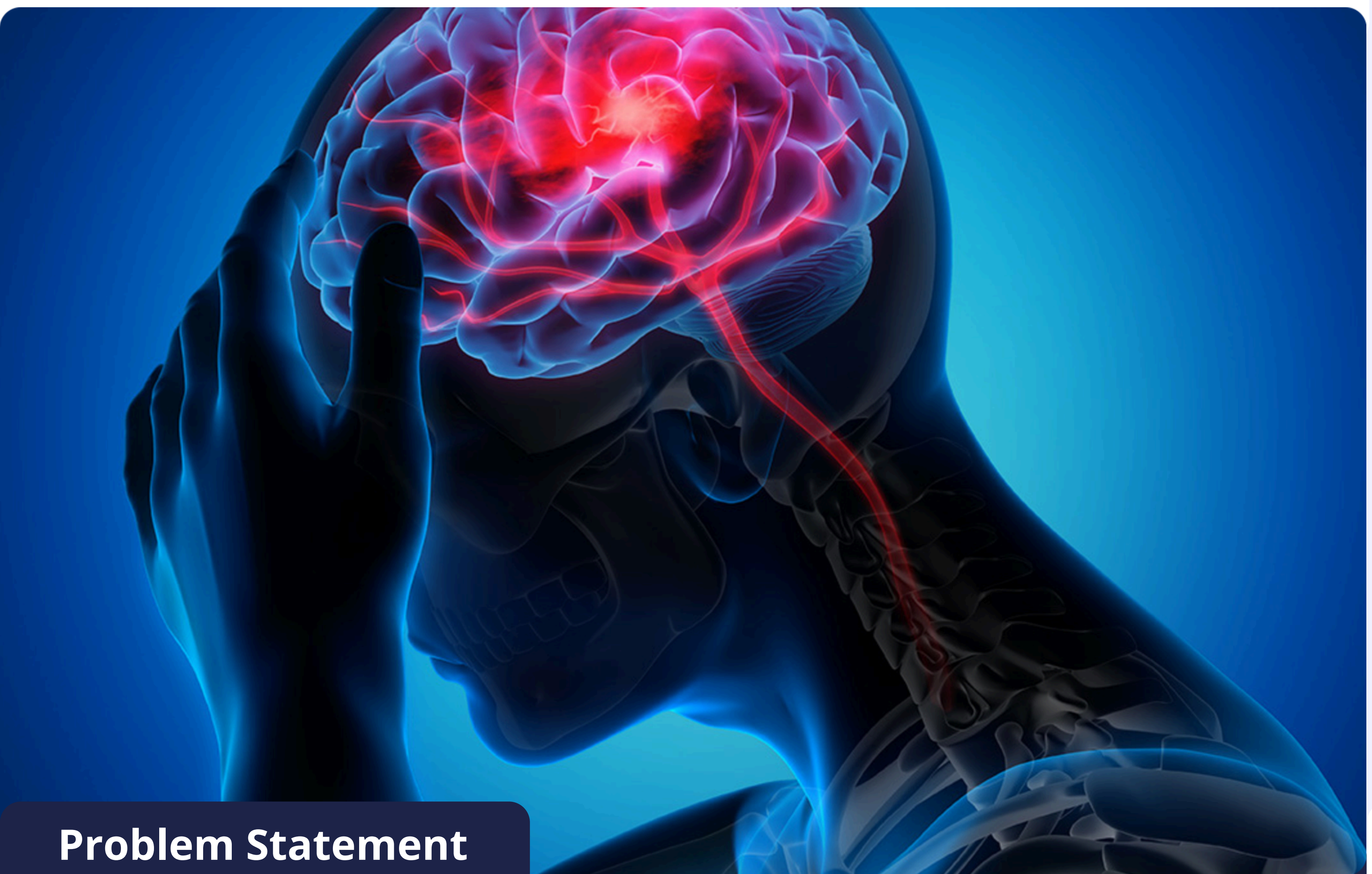
UNITED INTERNATIONAL UNIVERSITY

Team SuperNovas



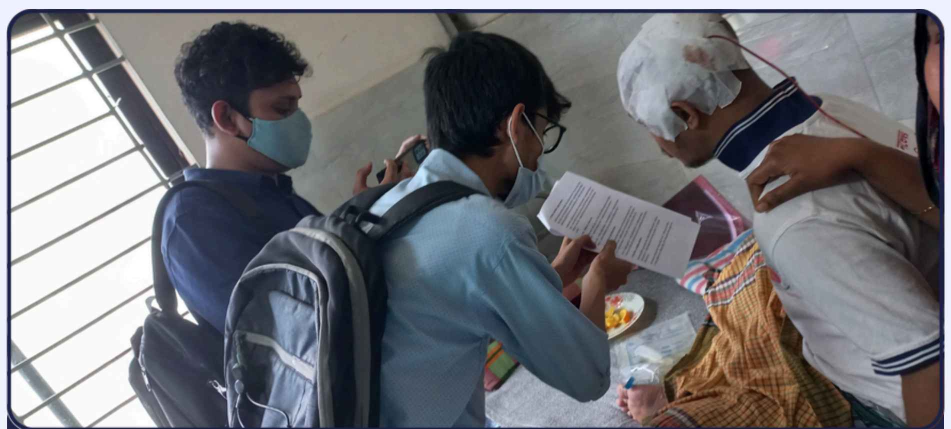
Multimodal Lip-Audio Fusion for Bengali Speech Reconstruction in Post-Stroke Patients

Problem Statement

Every year, nearly 12 million people worldwide suffer a stroke, including many in Bangladesh where prevalence is about 11.4 per 1,000. Post-stroke aphasia and dysarthria often severely impair speech, limiting communication. Current rehabilitation struggles when audio is weak, especially for low-resource languages like Bengali. A robust multimodal system combining lip movements and audio is urgently needed to restore intelligible speech and improve patients' quality of life.

Research Objectives

- Reconstruct Bengali speech using inputs of post-stroke patients.
  - আমি সবাইকে চিনতে পারছি
- Correct phonetic errors using context-aware language models.
  - আমি হাত খাই → আমি ভাত খাই
- Real-time system for practical use in clinical and daily cases.



Patient (Dyphasia): Sample No.1
Venue: Dhaka Medical Colleege

Data Collection

Component	Specification
Subjects	Strokepatients + healthy controls
Audio	A100microphone, 16 kHz, mono, WAV
Video	Frontal 720p, 30 FPS
Script	Bengalisentences (phoneme-rich)
Metadata	Age,gender, stroke score, etc.

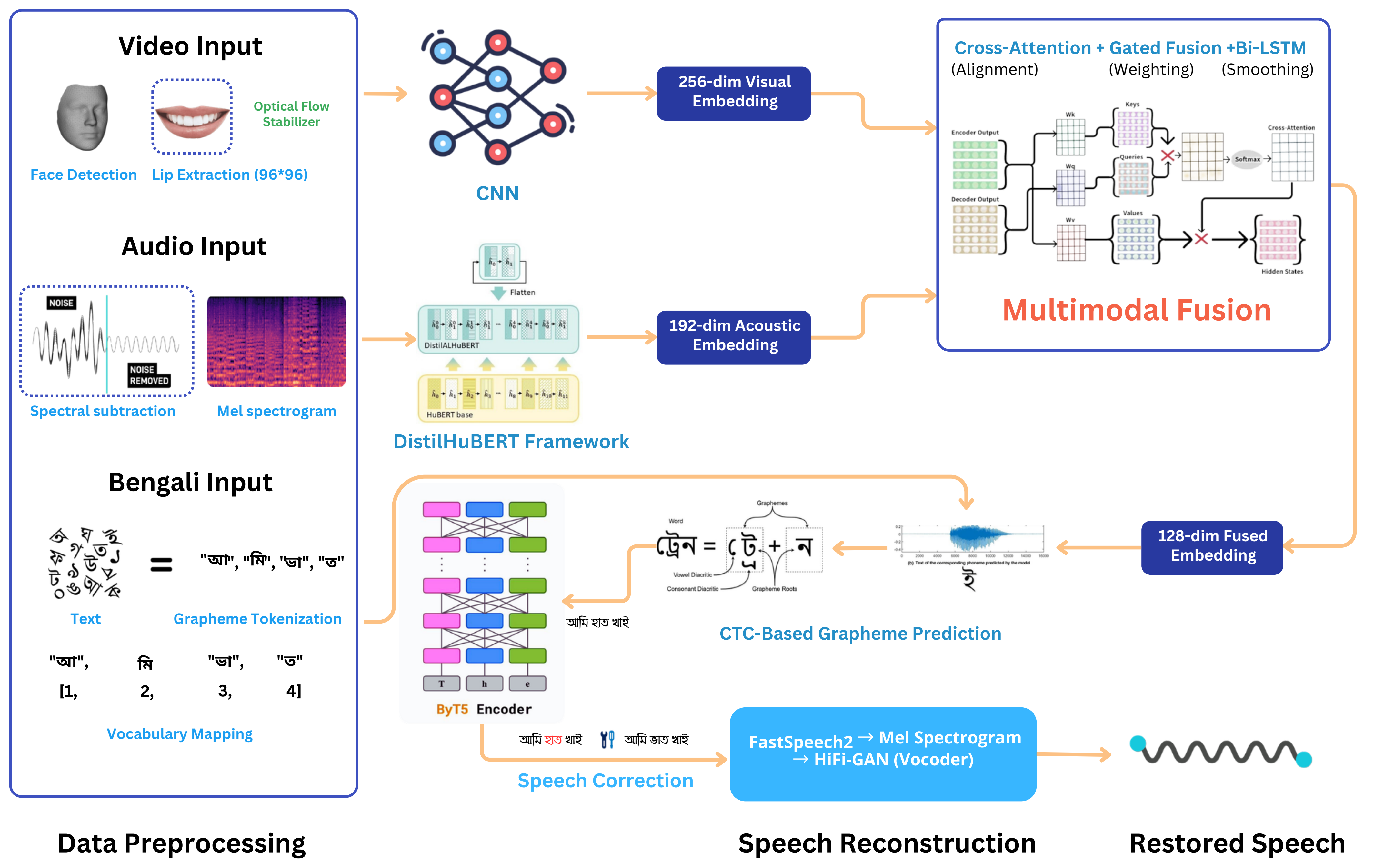
Ethical Considerations

- Consent Form
- Anonimity
- Doctor Supervision
- IRB Guidelines

Challenges

- Low Digital Literacy
- Ethical Consent
- Phonetic Complexity
- Patient Variability

Methodology



Research Gaps & Novelty

- Creation of a First-Ever Bengali Stroke Multimodal Dataset and Speech Reconstruction Pipeline
- Dynamic Gated Cross-Attention Fusion Model for Robust Speech Reconstruction
- LLM-Driven Grapheme-Level Correction Using ByT5 for Enhanced Clinical Usability

References

- Exploiting Audio-Visual Features with Pretrained AV-HuBERT for Multi-Modal Dysarthric Speech Reconstruction
- LipSound2: Self-Supervised Pre-Training for Lip-to-Speech Reconstruction and Lip Reading
- Indonesian Lip-Reading Detection and Recognition Based on Lip Shape Using Face Mesh and Long-Term Recurrent Convolutional Network
- Wearable Intelligent Throat Enables Natural Speech in Stroke Patients with Dysarthria

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